

Exercise 2.10 (1-6, 9-10)

1. The number x is positive, but the number y is not positive.

$$\sim (x > 0 \wedge y \leq 0) = \sim (x > 0) \vee \sim (y \leq 0) = (x \leq 0) \vee (y > 0)$$

x is not positive or y is positive

2. If x is prime, then $\sqrt{x}$ is not a rational number.

$$\sim ((x \text{ is prime}) \Rightarrow (\sqrt{x} \notin \mathbb{Q})) = (x \text{ is prime}) \wedge \sim (\sqrt{x} \notin \mathbb{Q}) = (x \text{ is prime}) \wedge (\sqrt{x} \in \mathbb{Q})$$

There exists some prime x where $\sqrt{x}$ is a rational number

3. For every prime number p , there is another prime number q with $q > p$.

Define $P[\mathbb{N}]$ as the set of all primes

$$\sim (\forall p \in P, \exists q \in P, q > p) = \exists p \in P, \forall q \in P, q \leq p$$

There exists some prime p such that for all primes q , $q \leq p$

4. For every positive number ϵ , there is a positive number δ for which $|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$

$$\begin{aligned} &\sim (\forall \epsilon \in (0, \infty), \exists \delta \in (0, \infty), (|x - a| < \delta) \Rightarrow (|f(x) - f(a)| < \epsilon)) \\ &\exists \epsilon \in (0, \infty), \forall \delta \in (0, \infty), \sim ((|x - a| < \delta) \Rightarrow (|f(x) - f(a)| < \epsilon)) \\ &\exists \epsilon \in (0, \infty), \forall \delta \in (0, \infty), (|x - a| < \delta) \wedge \sim (|f(x) - f(a)| < \epsilon) \\ &\exists \epsilon \in (0, \infty), \forall \delta \in (0, \infty), (|x - a| < \delta) \wedge (|f(x) - f(a)| \geq \epsilon) \end{aligned}$$

There exists some $\epsilon > 0$ such that for every $\delta > 0$, there is a number x where $|x - a| < \delta$ and $|f(x) - f(a)| \geq \epsilon$

5. For every positive number ϵ there is a positive number M for which $|f(x) - b| < \epsilon$, whenever $x > M$

$$\begin{aligned} &\sim (\forall \epsilon \in (0, \infty), \exists M \in (0, \infty), ((x > M) \Rightarrow (|f(x) - b| < \epsilon))) \\ &\exists \epsilon \in (0, \infty), \forall M \in (0, \infty), (x > M) \wedge (|f(x) - b| \geq \epsilon) \end{aligned}$$

There exists some positive ϵ such that for all positive M , there is a number $x > M$ where $|f(x) - b| \geq \epsilon$

6. There exists a real number a for which $a + x = x$ for every real number x .

$$\sim (\exists a \in \mathbb{R}, \forall x \in \mathbb{R}, a + x = x)$$

$$\forall a \in \mathbb{R}, \exists x \in \mathbb{R}, a + x \neq x$$

For every real number a , there exists a real number x such that $a + x \neq x$

9. If $\sin(x) < 0$, then it is not the case that $0 \leq x \leq \pi$.

$$\sim (\sin(x) < 0 \Rightarrow \sim (0 \leq x \leq \pi))$$

$$= (\sin(x) < 0) \wedge (0 \leq x \leq \pi)$$

There exists some value $x \in [0, \pi]$ such that $\sin(x)$ is negative

10. If f is a polynomial and its degree is greater than 2, then f' is not constant.

$$\sim (((f \text{ is a polynomial}) \wedge (\deg(f) > 2)) \Rightarrow (f' \text{ is not constant}))$$

$$= (f \text{ is a polynomial}) \wedge (\deg(f) > 2) \wedge (f' \text{ is a constant})$$

There exists a polynomial f such that the degree of f is greater than 2 and f' is a constant